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CASE STUDY

Off-label drug use – price analysis for Avastin® in ophthalmology

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Abstract

Purpose – The use of drugs outside the approved indication, called off-label use, is a growing phenomenon in medical practice. Especially, when a drug approved for systemic use is used locally in small quantities, the drug price per treatment can fall far below the potential value of the drug. This paper aims to outline the potential value of off-label Avastin® used for wet/neovascular age-related macular degeneration (AMD), the leading cause of legal blindness in western countries.

Design/methodology/approach – The van Westendorp price sensitivity measurement (PSM) method was used to find an optimal price range for an Avastin formulation potentially approved by German regulatory authorities for neovascular/wet AMD. A survey was conducted among a majority of German AMD specialists as the central part of the buying center. Germany, the largest pharmaceutical market in Europe, was selected for the study.

Findings – The effective total response rate was 25.1 percent (51 questionnaires).The price range found is clearly above the price for currently used "off-label" Avastin and far below the price for other comparable drugs approved for the treatment of neovascular AMD. The van Westendorp PSM method found for Avastin used in the eye the optimal price at €95 ($131), the indifference price at €200 ($276) and the acceptable price range between €90 ($124) and €310 ($428). The discussion provides further implications for pricing strategies for new drugs.

Originality/value – As price sensitivity and availability of drugs have become an increasingly political and public topic, off-label application of drugs is turning into an important challenge for the pharmaceutical industry. Taken in total, the results provide support for the view that off-label application not only creates major threats for pharmaceutical companies, but can also offer growth opportunities.

Keywords Pharmaceuticals industry, Drugs, Germany

Paper type Case study

Introduction

Age-related macular degeneration (AMD) is a chronic disease of the elderly population (de Jong, 2006). It is the most frequent cause of legal blindness in the western industrialized countries including Germany (Buch, 2005; Friedman *et al.*, 2004;

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Klaver *et al.*, 1998). Multiple studies have shown that the disease has a prevalence that increases with age (Friedman *et al.*, 2004). The most frequent cause for blindness among patients with AMD is the neovascular, so-called "wet" form of AMD. It is characterized by new, pathologic vessel growth in the eye.

The use of drugs outside the approved indication, so-called off-label use, is a growing phenomenon in medical practice. Especially, when a drug approved for systemic use is used locally in small quantities, the drug price per treatment can fall far below the potential value of the drug. This case study outlines the potential value of off-label use of Avastin® for wet/neovascular AMD.

Anti-VEGF therapy

Pathologic vessel growth, so-called angiogenesis, has been a major field of interest in cancer treatment since the early 1970s. Especially, one growth factor has been strongly associated with angiogenesis in different forms of cancers. This factor has been termed the "Vascular Endothelial Growth Factor" (VEGF). In the mid-1990s, the importance of VEGF slowly became also evident for a number of ocular diseases associated with pathologic vessel growth and increased vascular permeability (Aiello *et al.*, 1995; Spilsbury *et al.*, 2000; Ozaki *et al.*, 1997; Williams, 1998).

Anti-VEGF therapy was primarily developed to support chemotherapy in cancer treatment. Its value in ophthalmology became apparent later. Many of the current anti-VEGF drugs in ophthalmology are related anti-VEGF cancer drugs.

Several pharmaceutical companies have started to develop drugs to inhibit VEGF in the eye. The first companies to obtain regulatory approval for intraocular anti-VEGF treatment were Eyetech Pharmaceuticals (Pegaptanib/Macugen®-2004) and Genentech Inc. (Ranibizumab/Lucentis®-2006). Both drugs bind VEGF. Lucentis was designed to bind all subforms of VEGF, whereas Macugen® was designed to just bind the most frequent subform VEGF-165. Even though both drugs have not been directly compared in clinical trials, it appears evident, based on clinical trials and clinical experience, that Macugen is only able to slow down vision loss in patients with "wet" AMD, whereas Lucentis is able to stabilize vision in more than 90 percent of such patients at one and two years follow-up (Chakravarthy *et al.*, 2006; Rosenfeld *et al.*, 2006). In addition, a significant portion of patients is regaining vision. These are impressive results, especially when compared to treatment options ten years ago when destructive, thermal lasers were used to stop the progression of "wet" AMD with very poor functional outcomes.

Lucentis is widely used for "wet" AMD in countries where it is approved and reimbursed by health care systems.

Avastin

The worldwide first anti-VEGF drug to obtain marketing authorization was Avastin (Bevacizumab) in early 2004. It was first approved for the treatment of widespread cancer of the colon in combination with a chemotherapy regimen. The drug, designed for intravenous infusion, was derived from the same precursor mouse antibody as Lucentis. As of 2004, Avastin was used "off label" (outside the approved indication) first as a systemic infusion and later as an injection into the eye (Michels *et al.*, 2005; Moshfeghi *et al.*, 2006; Rosenfeld, Fung and Puliafito, 2005; Rosenfeld, Moshfeghi and Puliafito, 2005). There is an increasing number of prospective clinical studies

showing effectiveness of Avastin for "wet" AMD (Algvere *et al.*, 2008; Bashshur *et al.*, 2007, 2008; Schouten *et al.*, 2009; Weigert *et al.*, 2008). Large controlled clinical trials, financed without the support of the pharmaceutical industry, have just been initiated. The current literature places Avastin in the same range as Lucentis with regard to safety and effectiveness. Avastin, when used as an intraocular injection, is relatively inexpensive, since the drug has been priced for systemic infusions typically requiring 5 mg/kg Avastin (80 kg person: 400 mg/per treatment). For intraocular injection only a very small amount of Avastin® (commonly: 1.25 mg) is needed. In Germany, comparable to many other countries, the price for 1 mg Avastin is currently about €4.40. Figure 1 shows price and effectiveness for different treatment options in neovascular AMD.

The splitting of the drug under sterile conditions by a pharmacy is more expensive than the Avastin drug dose itself. This procedure is necessary because Avastin is marketed in 100 and 400 mg vials. The cost of the preparation is variable but commonly ranges between €20 and 30. Owing to similar safety and effectiveness to Lucentis and due to the very low price per ocular treatment, Avastin has become a worldwide used "off-label" therapy for "wet" AMD. Avastin is also being used with increasing frequency for other very common diseases such as diabetic retinopathy/maculopathy or retinal vascular occlusions.

Limits for "off-label" therapy in Germany

In the current situation, it appears obvious that the Lucentis is the drug most AMD-specialists would use for the treatment of wet AMD. This drug is effective, available, and is usually reimbursed by health care insurances. The default decision seems clearly in favor of Lucentis. However, a second look shows that the situation is more complex.

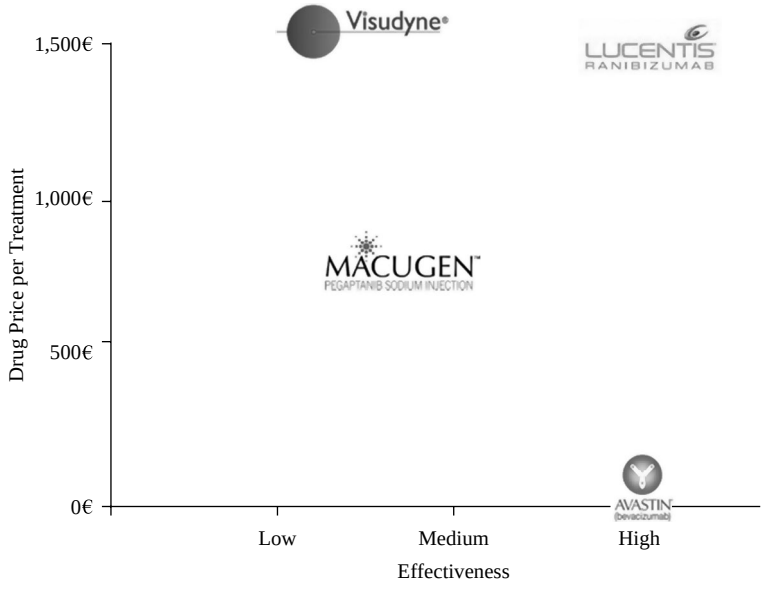


Figure 1.
Price and effectiveness of different drugs/treatments for neovascular AMD

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There is strong cost awareness in Germany and overall budgets are limited (Statistisches Bundesamt, 2006). Even though Lucentis is reimbursed, it is obvious to most ophthalmologists that the introduction of an inexpensive new treatment will have significant impact on future budgets and reimbursement of other ophthalmologic services in private practice and in clinics. In addition, there is the option for inexpensive but "off-label" Avastin. AMD specialists have used this drug "off-label" for more than a year before Lucentis was approved and are currently using it for VEGF-related eye diseases other than neovascular/"wet" AMD, as mentioned earlier.

The German legislature has clear guidelines for the reimbursement of off-label drugs. Three aspects have to be fulfilled to obtain reimbursement for off-label therapy:

(1) treatment of a severe disease;

(2) no other approved therapy available; and

(3) good chance of treatment success based on current data.

With the approval of Lucentis, an effective drug for all lesion types of neovascular AMD has become available, which should lead to a reimbursement halt for Avastin in Germany. However, it currently becomes obvious that this legislation is not enforced. A further important aspect is the potential risk of legal actions against AMD specialists following an adverse event using Avastin for the treatment of neovascular AMD. To date, no such case has been reported.

The present study addresses several areas of interest. Different from pricing finding strategies for new drugs under patent protection used by the pharmaceutical industry, the study uses – as commonly done for products in unregulated markets – the central part of the buying center to find a price range for a potentially approved Avastin eye formulation for the treatment of wet AMD in Germany.

Several aspects affect the industry when considering prices for innovative, new drugs. On the one hand, there are clearly drivers for lower drug prices such as overall health care costs, rising health care insurance costs, decreasing budgets for ophthalmologists, and risk for lower reimbursement of medical professional work in case of higher drug prices. On the other hand, very low prices for new, innovative, and most effective drugs will clearly limit pharmaceutical industry's interest in further research and development of new products in that field, and will subsequently affect future treatment options. In addition, a marketing authorization improves doctor's safety with regard to reimbursement and legal issues and justifies a higher drug price.

The purpose of this case study is to discuss the pricing implications for an Avastin formulation potentially approved in Germany in comparison to other treatments for wet AMD.

Methods

Germany was selected for the Avastin price analysis in ophthalmology for several reasons. The German market is the largest pharmaceutical market within the European Union and the third largest in the world. German AMD specialists, different from their colleagues in other EU countries, had in 2007 the largest experience with all anti-VEGF drugs. This was primarily due to the rapid adoption of Avastin as off-label AMD treatment and the fast nation-wide reimbursement of Lucentis following its European approval. In addition, cost-consciousness is high among German ophthalmologists in the context of overall spending in ophthalmology being

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reduced over the last years (−13 percent between 2002 and 2004; Statistisches Bundesamt, 2006).

The AMD specialists were considered as the central part of the buying center (Figure 2), in an intermediary role among patients, pharmaceutical industry, insurance companies, and health care authorities.

AMD specialists, familiar with AMD treatments, were specifically addressed in order to obtain the most valuable answers and to outline the current market situation. The selectivity for this specific subgroup of ophthalmologists clearly limits the size and the power of the survey. The publicly available PDT/Visudyne® user list (PDT, 2007), last updated January 2007, was used to contact AMD specialists in Germany. This list has the advantage that it is the most comprehensive publicly available list of AMD specialists belonging to different interest groups (e.g. private practice vs public hospital). In total, 203 surveys were sent to AMD specialists.

In general, it is most difficult to obtain a precise product price using a survey since the customer-buying center typically has difficulty naming an exact price. The customer buying center rather buys a product within a specific price range. van Westendorp took this aspect into account by developing the price sensitivity measurement (PSM) method, which is a technique developed as a survey methodology for the measurement of price perceptions (Vlosky *et al.*, 1999). The PSM has been adapted by researchers and has been delivering reliable research results for specific product contexts (Allen and Maybin, 2004; Lyon, 2002; Draeger, 2000).

The van Westerdorp method allows obtaining an acceptable price range for a certain product. A majority of customers accept a price within this range. The most relevant findings in this type of analysis are considered to be the optimal price point (OPP), the indifference price point (IPP) and the acceptable price range. To determine the price sensitivity, the van Westendorp method requires answering a special set of four questions (Allen and Maybin, 2004; Draeger, 2000). To derive a high-external validity, the structure of the individual questions asks the respondents to qualify their price based on an implied assumption about quality (Vlosky *et al.*, 1999). Furthermore, the researcher forces a response with no point of reference other than the concept and the question (van Westendorp, 1976). The four questions, which directly ask respondents about prices are:

(1) Up to which price would you consider the product as low priced?

(2) At what price would you say the product is expensive but worth a consideration?

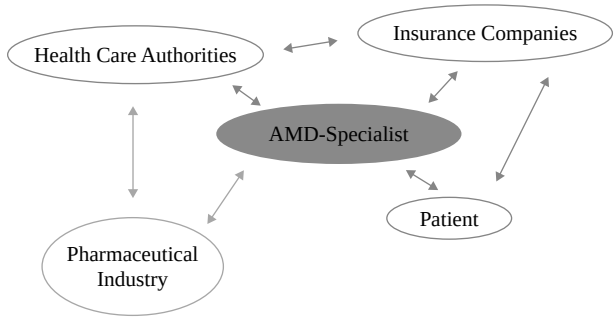


Figure 2.
Buying center for AMD treatments

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(3) At what price would you consider the product as too expensive and you would stop buying it?

(4) Below which price would you consider the product too cheap and you would doubt its quality?

The intersection of the cumulative frequency distribution of the questions "too expensive" and "too cheap" is the OPP. At this point, the resistance to buy a product is the lowest. Optimal in this sense refers to the equal trade-off in extreme sensitivities to the price at both ends of the price spectrum. The IPP is found at the intersection of the cumulative frequency distribution of the questions "expensive" and "low priced". It refers to the price point with an equal number of respondents rating the price as either "low priced" or "expensive". The acceptable price range of a product is found between the crossing of the curves "too expensive" and "low priced" and "too cheap" and "expensive".

We integrated the van Westendorp PSM method for an Avastin formulation potentially approved in Germany for the treatment of "wet" AMD into a larger questionnaire.

The price for Avastin when used intraocularly is calculated based on the price of €4.40/mg ($6/mg). The injection procedure (filling of dead space in injection needle) requires l2.7 mg in order to inject 1.5 mg. Accordingly, the Avastin drug price for a single intraocular treatment was set at €12 ($16.60). The most variable price of compounding the 400 mg Avastin vials is not taken into calculation. This compares to a Lucentis price of about €1,500 ($2,070) per treatment.

To position the price analysis in a relevant context, we additionally evaluated baseline demographics (age and gender), the setting in which AMD patients were treated (public hospital, private practice, and private hospital), the number of AMD treatments performed each month, and the use of certain treatments.

Sample description

A total of 203 questionnaires were sent to AMD specialists listed in the publicly available PDT/Visudyne user list in June 2007. In total, 51 questionnaires were returned by August 15, 2007. The response rate was 25.1 percent.

The mean age of the responding AMD-specialist was 44.2 years (SD 8.2 years). The majority was male (40 vs 11).

The mean number of patients with neovascular AMD treated per month per AMD specialist was 117. Based on these data, the participating AMD specialists would perform approximately 71,600 treatments per year. Taking into consideration a minimum (van Leeuwen *et al.*, 2003) and maximum (Schrader, 2006) estimated incidence of neovascular AMD per year in Germany, the market share of each AMD treatment, the mean number of treatments of each AMD treatment per year (Bressler, 2001; D'Amico *et al.*, 2006; Fung *et al.*, 2007; Gragoudas *et al.*, 2004; Leydolt *et al.*, 2008), and an estimated treatment duration of two years (except Lucentis – was approved February 2007), an estimate on the overall number of AMD treatments within one year can be given.

The estimated range of 20,000-50,000 new wet AMD cases per year in Germany would lead to approximately 137,000-345,000 AMD treatments per year. The total

proportion of wet AMD treatments performed by AMD specialists responding to the survey was between 20.75 and 52.14 percent.

Market share

According to the questionnaire, the shares for different treatments for neovascular AMD in Germany were distributed in July 2007 as follows: Lucentis was used on average in 57.1 percent cases for treatment in wet AMD, Avastin in 29 percent, Macugen in 5.9 percent, PDT with Visudyne in 5.5 percent and other treatments in 2.3 percent cases. Figure 3 shows these results.

Based on the market share for each drug currently used for the treatment of wet AMD, sales can be estimated for 2007 as shown in Table I.

Price analysis

The van Westendorp analysis provided the OPP, the IPP and the acceptable price range for an Avastin formulation potentially approved for wet AMD. Figure 4 shows the cumulative curves for each of the four questions.

The OPP is at the intersection of the cumulative frequency curves "too cheap" or "too expensive". The resistance "to buy" is here the lowest. The OPP is, according the survey at €95 ($131).

The IPP refers to the price point with equal number of respondents rating the price as either "low priced" or "expensive". The IPP according to the survey was at €200 ($276).

The upper and lower limits of the acceptable price range are found at the crossing of the curves "too expensive" and "low priced" and "too cheap" and "expensive". The acceptable price range was found to be between €90 ($124) and €310 ($428).

Table II estimates the sales for an Avastin formulation potentially approved for wet AMD based on OPP, IPP, and maximum price within the acceptable price range.

Discussion

The current price range of about €12 ($16.60) for Avastin to €1,500 ($2,070) for Lucentis for a single intraocular treatment using two very similar anti-VEGF drugs is currently difficult to comprehensively explain. The strongest arguments in favor of a higher

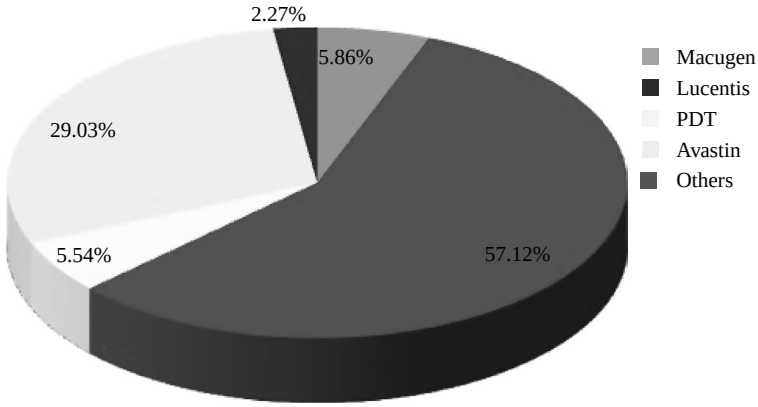


Figure 3.
Market share of AMD treatments according to survey in July 2007 in Germany

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Table I.
Estimated sales for 2007 based on market share in survey

	No. of patients started TX in 2007	No. of patients started TX in 2006	Mean number of TX (1st and 2nd year)	Drug price (€)	Estimated total sales (€) 2007 (Mio)
Lucentis®					
Min	11,400	0	5.6 (1st year)[a]	~ 1,500	~ 96
Max	28,500	0			~ 239
Macugen®					
Min	1,200	1,200	8.5 (1st year)	~ 800	~ 15
Max	2,900	2,900	7 (2nd year)[b]		~ 36
Visudyne®					
Min	1,100	1,100	3.4 (1st year)	~ 1,600	~ 10
Max	2,800	2,800	2.2 (2nd year)[c]		~ 25
Avastin®					
Min	5,800	5,800	4.3 (1st year)	~ 12 (2.7 mg from 400 mg vial)	~ 0.6
Max	14,500	14,500	4.3 (2nd year)[d]		~ 1.5

Notes: [a]PrONTO Study (Fung *et al.*, 2007); [b]vision studies (Gragoudas *et al.*, 2004; D'Amico *et al.*, 2006; Chakravarthy *et al.*, 2006); [c] treatment of Age-related Macular Degeneration with Photodynamic Therapy (TAP) Study Group (1999; Bressler, 2001); [d]no 2 year data available (Leydolt *et al.*, 2008). TX, treatment; The table's estimates contain several assumptions: incidence of neovascular AMD per year is estimated for Germany based on the Rotterdam Eye Study (min) (van Leeuwen *et al.*, 2003) and on estimates by Schrader (max) (Schrader, 2006). Number of retreatments in 1st and 2nd year of treatment is based on selected studies published in the medical literature; patients requiring more than two years of treatment are not taken into account; loss of patients in follow-up is not taken into account

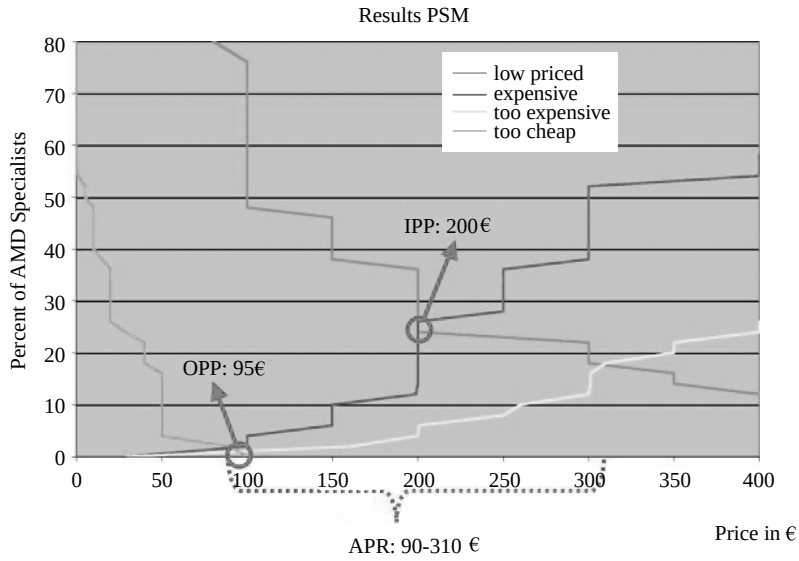


Figure 4.
van Westendorp PSM

Lucentis price are the higher level of scientific evidence and its approval for intraocular use in wet AMD. The growing political debate on this topic in Europe and the United States obviously indicates that these arguments are not satisfactory to everyone (Lee and Radford, 2008; Pollack, 2005). The presented pricing strategy for an Avastin formulation approved for intraocular use in neovascular AMD fundamentally differs from the price evaluation strategies used by the pharmaceutical industry for newly approved drugs under patent protection. Drug prices are most commonly based on costs for research and development, production, marketing and sales plus two variables: profit margin and accruals for the company's unsuccessful drugs that never reached the market. The pharmaceutical industry estimates the cost of bringing a new drug to the market currently at €800 million ($1.1billion). Profit margins can be estimated, but accruals for unsuccessful drugs are impossible to estimate. For market leaders and blockbusters like Lucentis or Avastin (cancer treatment) these accruals are above average.

The concept of presented price analysis is based on the finding that the market of anti-VEGF drugs in ophthalmology is more competitive than expected. The off-label status of a drug regarding a specific indication is a limiting factor but clearly does not exclude a drug from the market. In this market situation, the AMD specialists is seen as the central part of the buying center and as the mediator within the network of patients, pharmaceutical industry, insurance companies and health care authorities. Based on these findings the AMD specialists are addressed in the price analysis.

To find an acceptable price range from the perspective of AMD specialists, the van Westendorp PSM method was chosen. This tool is commonly used by marketing specialists. It aims to find an acceptable price range rather than a specific price for a product in open, unregulated markets. The OPP for the ophthalmic Avastin formulation, the point where the resistance to buy is the lowest, was at €95 ($131). This is about eight times the current Avastin price when used off-label. Estimated sales in

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Table II.
Estimated sales (2007) for Avastin® formulation potentially approved for wet AMD

Avastin®	No. of patients started TX in 2007 Min	Max	No. of patients started TX in 2006 Min	Max	Mean number (per year) of TX 1st and 2nd year	Drug price (€)	Estimated total sales (€) 2007 (Mio)
Base case	5,800	14,500	5,800	14,500	4.3	~ 12 (3 mg from 400 mg vial)	~ 0.6 ~ 1.5
OPP (29 percent market share)	5,800	14,500	5,800	14,500	4.3	95	~ 4.7 ~ 12
OPP (58 percent market share)	11,600	29,000	5,800	14,500	4.3	95	~ 7.1 ~ 18
IPP (29 percent market share)	5,800	14,500	5,800	14,500	4.3	200	~ 10 ~ 25
IPP (58 percent market share)	11,600	29,000	5,800	14,500	4.3	200	~ 15 ~ 29
MAX. PR (29 percent market share)	5,800	14,500	5,800	14,500	4.3	310	~ 16 ~ 39
MAX. PR (29 percent market share)	11,600	29,000	5,800	14,500	4.3	310	~ 23 ~ 46

Notes: OPP, optimal price point; IPP, indifference price point; MAX. PR, maximum of acceptable price range; TX, treatment; maximum and minimum number of new cases with neovascular AMD in Germany per year

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Germany based on the OPP could reach about €16 million ($22 million) per year, in case of an incidence of wet AMD of 50,000 per year and a doubling of the current market share. The IPP according the survey was at €200 ($276). Transferring assumptions for incidence of wet AMD and market share from the OPP to the IPP would lead to sales of €29 million ($40 million) per year. The acceptable price range was found to be between €90 ($124) and €310 ($428). At the upper limit of the acceptable price range, the point of marginal expensiveness, sales would increase to €45 million ($62 million) under the conditions described above for OPP and IPP.

The price range resulting from the presented study can be interpreted from different perspectives. On the one hand, it reflects the value of such therapy from the perspective of an AMD specialist. On the other hand, it could show the monetary value of reducing legal costs and reimbursement risks to AMD specialists by approving Avastin for wet AMD. But, most likely it is a price compromise taking into account interests of all parts of the buying center.

Major disadvantages of the van Westendorp PSM method used in the survey are according to the literature inconsistent answers to the four questions in up to 20 percent. In addition, the price focused questions could lead to too low prices (Lyon, 2002). To limit especially the underestimation of prices monadic tests asking a single buy (yes or no) question are usually favored. However, monadic testing requires a relatively narrow price focus and a large number of interviewees. Neither is the case in the presented situation. The price range of comparable anti-VEGF drugs ranges from €12 ($16.60) to €1,500 ($2,070) and the overall potentially accessible number of AMD specialists in Germany was limited. Following the van Westendorp PSM a monadic testing could be performed based on the findings in this survey and within a larger international community of AMD specialists.

The study has further limitations. There is lack of precise data on the number of new wet AMD cases in Germany per year, requiring estimates on a maximum and minimum incidence. The number of treatments per year for each AMD drug is based on selected studies and may not reflect daily clinical routine. Furthermore, patients lost in follow-up and requiring treatment for more than two years are not taken into account.

In the longer term, an approved Avastin formulation for the intraocular treatment of wet AMD could be of advantage for all stakeholders. More patients, especially when taking a broader, international perspective, would gain access to an effective anti-VEGF therapy, not only for the treatment of neovascular AMD, but potentially also for the treatment of other frequent neovascular/exudative ocular diseases such as diabetic retinopathy/maculopathy or retinal vessel occlusions.

It is difficult to predict the outcome of a comparable survey in the USA, the largest pharmaceutical market in the world. There is however some rationale that the price range for an Avastin formulation potentially approved for the treatment of "wet" AMD by the FDA would be lower. Reasons for this assumption are the large percentage of AMD specialist in the USA using Avastin for "wet" AMD, the reimbursement of Avastin for "wet" AMD by Medicare and the large "Comparison of Age-related macular degeneration Treatment Trials" comparing directly Avastin to Lucentis. Genentech's efforts to stop the intraocular use of Avastin by "Dear doctor letters" indicating that Avastin is dangerous to be used in the eye and by threatening compounding pharmacies (splitting of Avastin into small doses) with Avastin delivery

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stoppages, are likely to reduce the price range. The authors consider scenarios based on the current situation:

- The pharmaceutical industry decides to pursue an approval process of Avastin for neovascular AMD in Germany and other countries outside the USA. It is likely that a slightly modified Avastin formulation will be proposed to justify an increase in sales price. It appears possible that an Avastin formulation for ophthalmic use could be sold for €200-310 ($276-428) according to the survey. Sales in Germany could be at the range of €40 million per year. Regulatory agencies could potentially accept an accelerated approval process not only for neovascular AMD but also for other neovascular/exudative ocular diseases. The sales price for Lucentis will be reduced to maintain a relevant market share.
- The pharmaceutical industry decides against an approval process for Avastin for the treatment of neovascular AMD. Several European countries will initiate an approval process against the interest of the industry. In Italy, such an approval process has already taken place. The German Minister of Health has considered such a process. She is gaining support for this strategy from health care insurance companies. Owing to the low price (€12/$16.60), sales of Avastin for ophthalmic use will be in the range of €1.6 million ($2.2 million) in Germany per year. Lucentis will almost disappear from the market. The future incentive for the pharmaceutical industry to bring new drugs for neovascular AMD into the markets will be reduced.
- Neither the pharmaceutical industry nor governmental agencies pursue an approval process for ophthalmic use. This represents more or less the current situation. The use of Avastin will depend on further trials comparing Avastin and Lucentis, on reimbursement strategies by insurance companies and potentially on court rulings. If the situation in Germany is similar to the USA, Lucentis sales will slowly decrease whereas Avastin sales will increase. Sanctions against German compounding pharmacies by the pharmaceutical industry will have – as in the USA – little effect.

The price setting strategy, presented in this case may be a useful way to find an acceptable price range for a new, innovative drug in a presumably regulated, but in reality competitive market with large price spreads.

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